



# FoodieClock: Smart Fridge Companion

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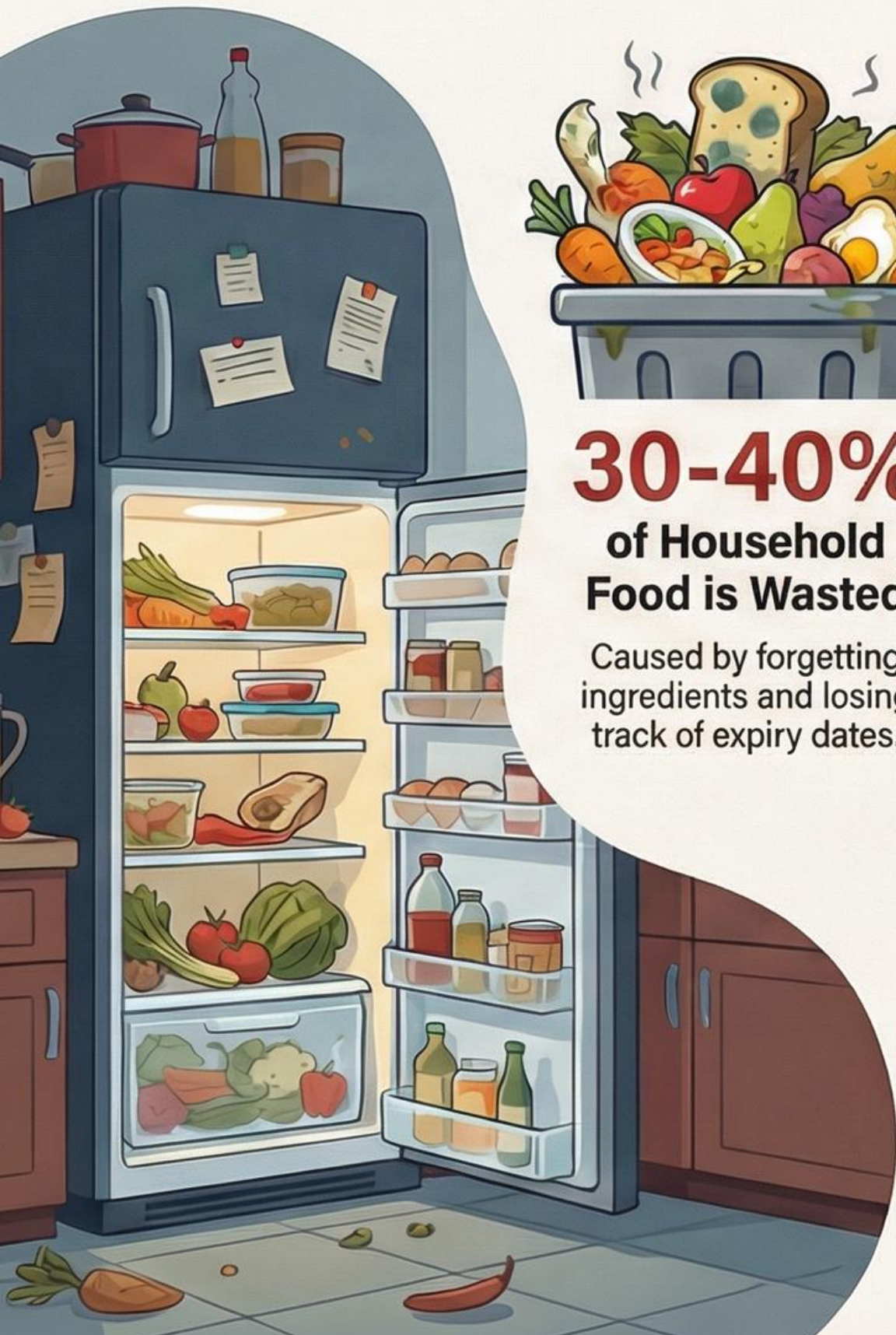
JIANG, Liuyue  
SUN, Ying  
GUO, Steven

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# Meet Your Smart Fridge Companion

## The Problem: Kitchen Waste & Indecision



**30-40%**  
of Household  
Food is Wasted

Caused by forgetting  
ingredients and losing  
track of expiry dates.



**Manual Tracking  
is Ineffective**

It's tedious, inconsistent,  
and rarely kept up-to-date.



**The Daily "What to  
Eat?" Struggle**

It's hard to find recipes that  
match the ingredients you  
already have.

## The Solution: An AI-Powered Assistant



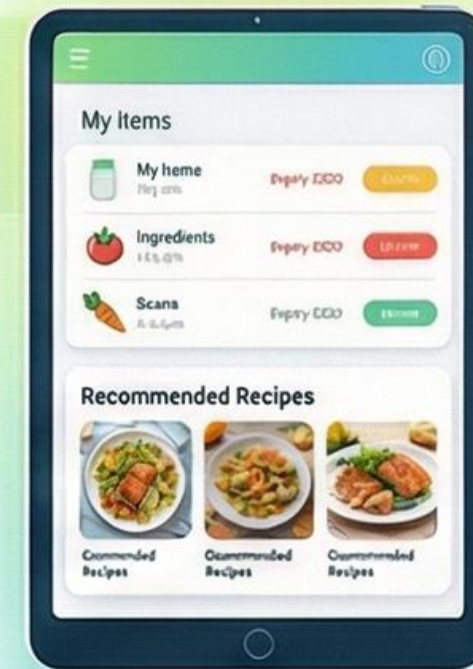
**How It Works  
in 3 Simple Steps**



**1. Scan/Input  
Ingredients**

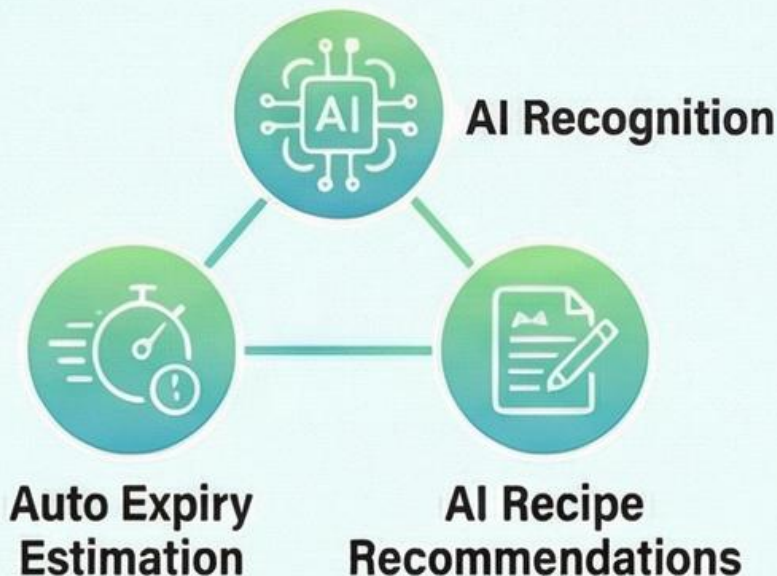


**2. AI identifies food,  
estimates expiry,  
and finds recipes.**



**3. View your  
dashboard.**

## Key Features



**The Impact: Save Money &  
Simplify Meals**

Cut food waste, solve the 'what to eat'  
problem, and organize your kitchen.



# Target Audience

**01**

## **Urban office workers**

Fast-paced, often stockpiling ingredients, afraid of expiration and waste, need efficient food management tools;

**02**

## **Kitchen novice**

Lacking experience in ingredient combination, they need convenient cooking guidelines for "doing whatever is available".

**03**

## **A sophisticated young person living alone**

Responsible for the family's diet, needs to precisely control the freshness of ingredients and ensure the safety of children's food.

**04**

## **Mother/Housewife**

Pays attention to a healthy diet, stocks up in small quantities to avoid waste, and needs inspiration for personalized recipes.





# Never Waste Food Again: The FoodieClock Smart Kitchen Journey

## 1. Discovery & Effortless Onboarding



## 2. The Core Loop: Smart Management in Action



## 3. Driving Loyalty & Advanced Use





# Core Use-Cases & main function

| Use-Case                 | Key Scenario                        | Core Steps                                                   |
|--------------------------|-------------------------------------|--------------------------------------------------------------|
| 1. Intelligent Entry     | Quick ingredient inventory addition | 1. Photo (OCR/AI) /manual/voice input<br>2. Confirm & save   |
| 2. Expiry Reminder       | Avoid food waste                    | 1. Auto-set triggers (3d/1d/expiry)<br>2. Push notifications |
| 3. Recipe Recommendation | Use expiring ingredients            | 1. Sync inventory<br>2. Filter & view recipes                |



# System Architecture

The system is designed to track food items in the refrigerator and their expiration dates.



**1. CAPTURE:** The user takes a photo on their device.

The iOS (Swift) or Web (React/Vue) app sends the image to the backend.

**2. ANALYZE:** AI services extract the details.

A third-party OCR service reads the ingredient name and expiration date from the image.

**3. STORE:** Information is saved and synced.

Data is stored in the central database (MySQL) and made available across devices.

**4. SCHEDULE:** An alert is automatically set.

The backend Expiry Reminder Service schedules a future push notification.

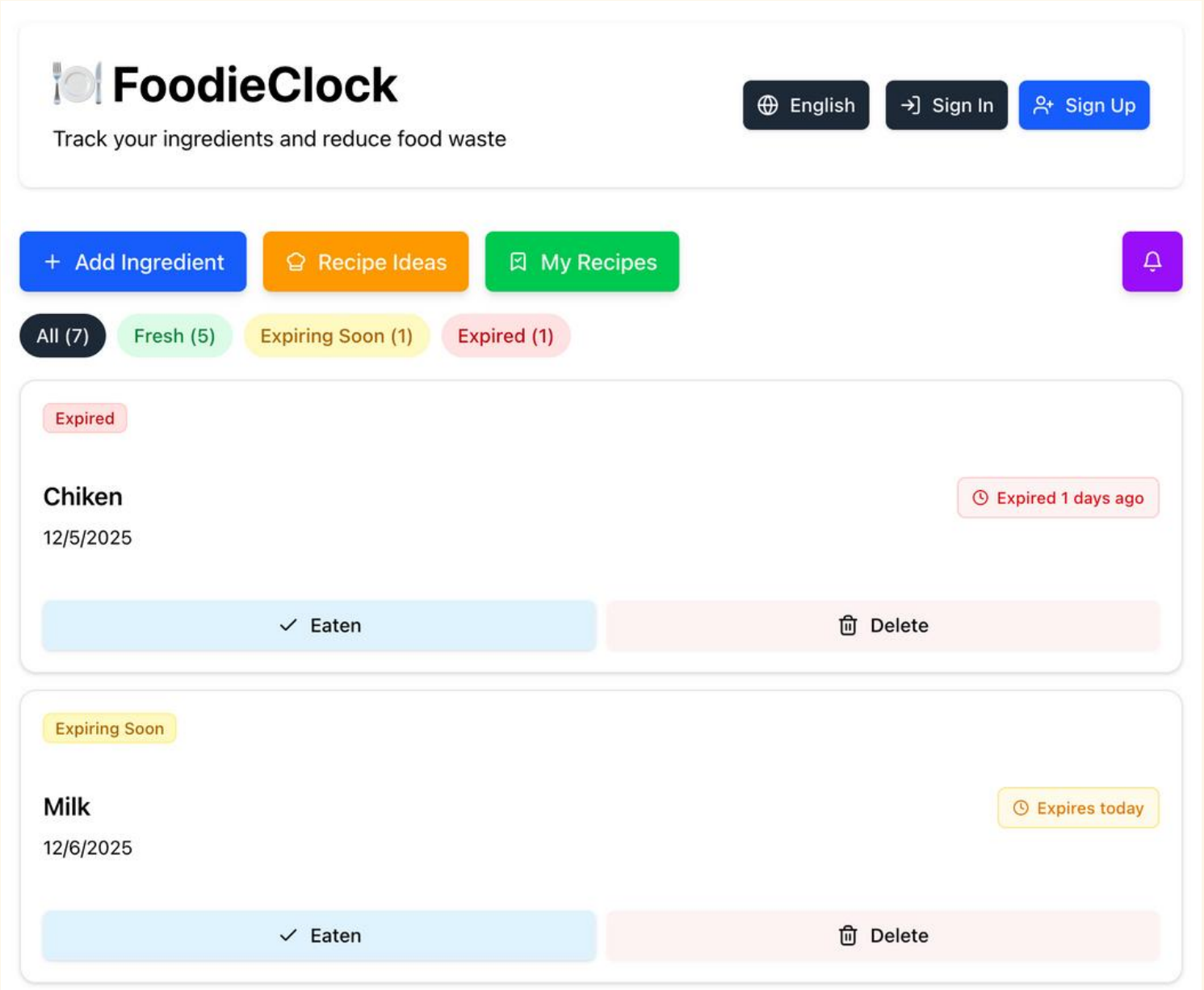
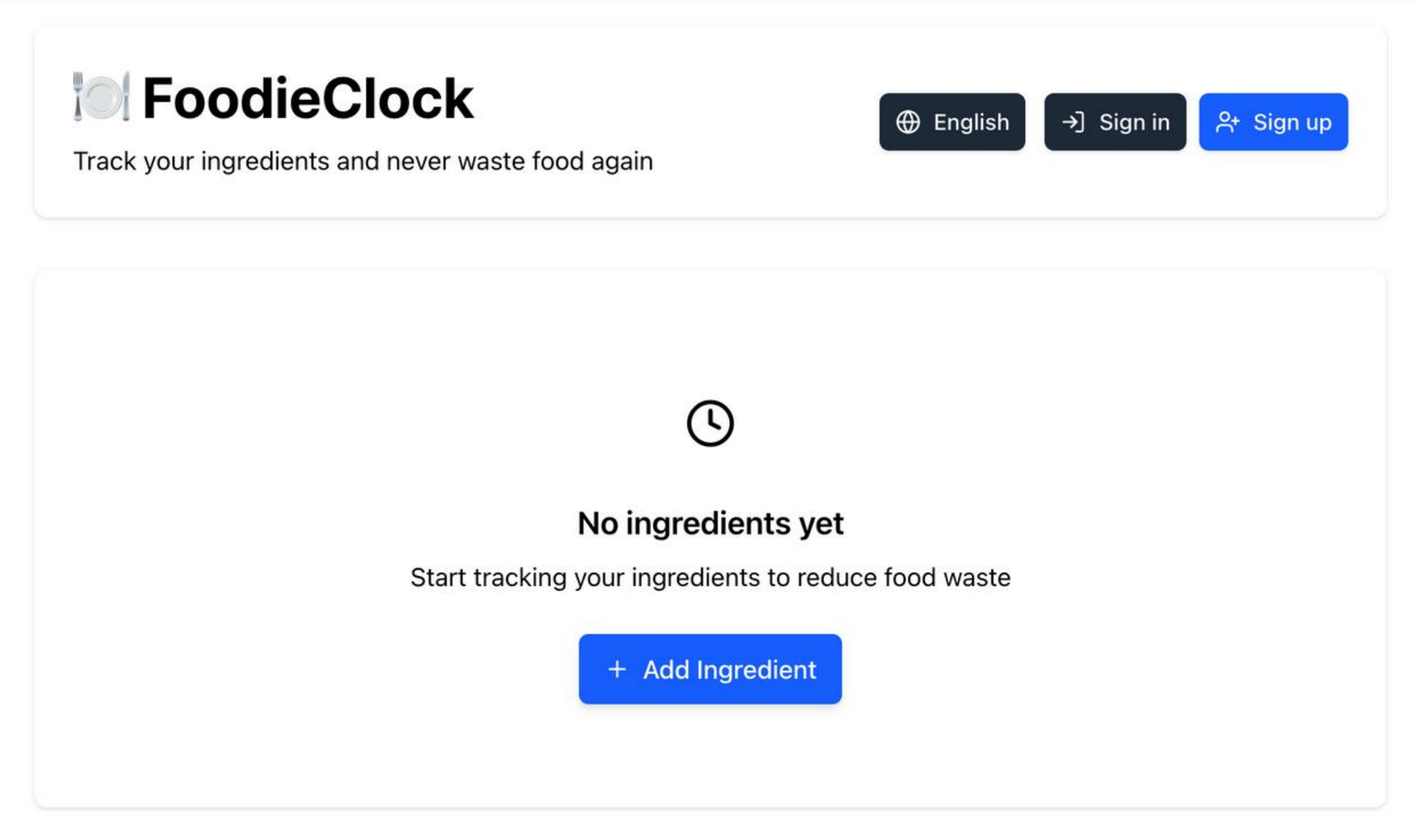
**5. NOTIFY:** The user gets a timely reminder.

A push notification (APNs/Web Push) is sent before the food expires.



# Web-end Interface

## Main Pages — sign up/in, select language



# Web-end Interface

×

## Add New Ingredient

Choose how you want to add your ingredients

Manual

Image

Audio

Ingredient Name

e.g., Milk, Eggs, Chicken

Life Span (Days)

7

Expiry Date

2025/12/08

📅

Add Ingredient

**3 input methods**

×

## Add New Ingredient

Choose how you want to add your ingredients

Manual

Image

Audio

📷

Upload an image of your ingredients

AI will detect ingredients and suggest expiry dates

📁 Choose File

📷 Take Photo

AI will detect ingredients and suggest expiry dates

×

## Add New Ingredient

Choose how you want to add your ingredients

Manual

Image

Audio

Language:

English

▼

Start Recording

Tap the button and speak your ingredients with their expiry dates.  
AI will automatically detect them.



# Web-end Interface

## Notifications

✕

⚙️ Reminder Settings

🔔 Push Notification

Enable Push Notifications  
Receive browser notifications about expiring ingredients

🔔 Test Notification

🕒 Notification Time

Reminder Time08:00

Custom Advance Notice

14 days before expiry

Get notified this many days before expiry

Save Settings

⚠️ Expire Reminder

✕

Milk

Today

Cilantro

4 days left

Shrimp

7 days left

Chicken

7 days left

Abalone

7 days left

Close



# Web-end Interface

## Recipe ideas

### Recipe Ideas

Plan for how many days  
1 day

Breakfast dishes  
1 dish

Lunch dishes  
1 dish

Dinner dishes  
1 dish

Dietary Preferences (Optional)

Quick & Easy

Healthy

Vegetarian

High Protein

Kid-Friendly

Low Carb

Budget-Friendly

Prioritize Soon-to-Expire

Other requests, e.g., no spicy, less oil, more vegetables...

Generate Recipes

Will generate meal suggestions based on your 7 ingredients

### Recipe Ideas

Generate New Plan

Regenerate All

Tue  
2

Tuesday, Dec 2

Breakfast

BREAKFAST

Scrambled Eggs

Fluffy scrambled eggs cooked to perfection.

Prep: 5 minutes

Cook: 5 minutes

Lunch

LUNCH

Egg and Milk Soup

A simple and creamy egg soup.

Prep: 5 minutes

Cook: 10 minutes

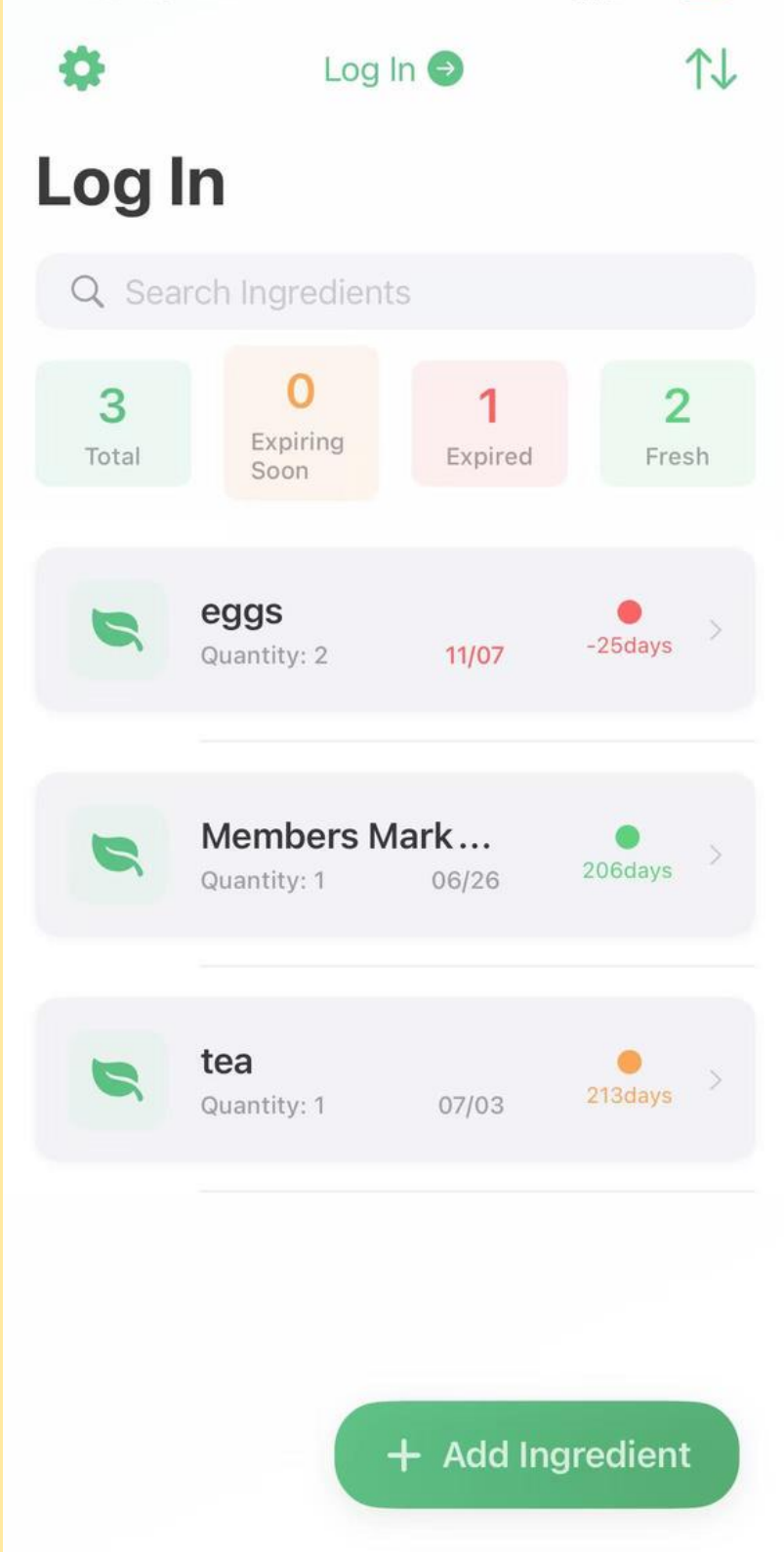
Dinner

DINNER

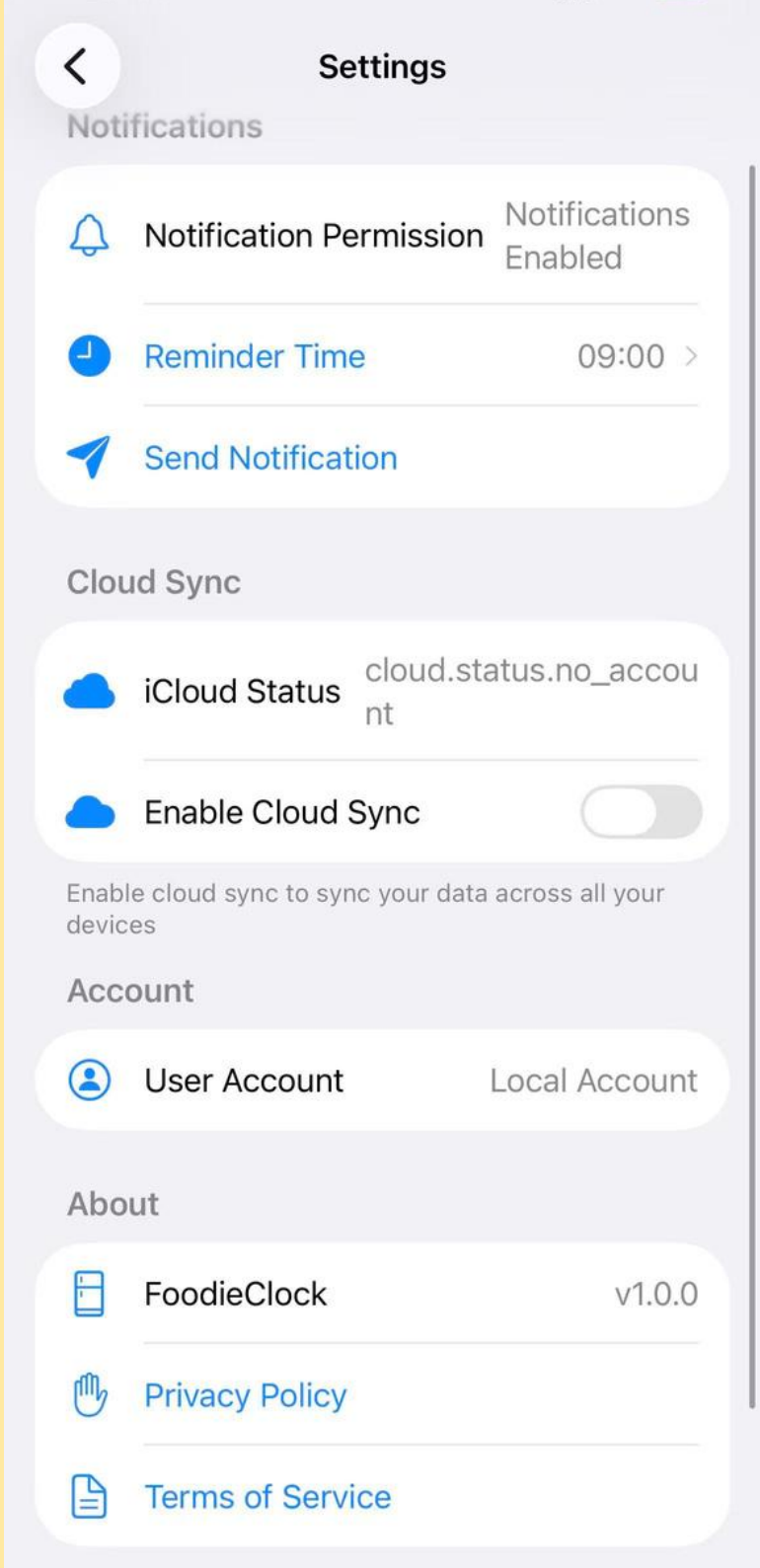
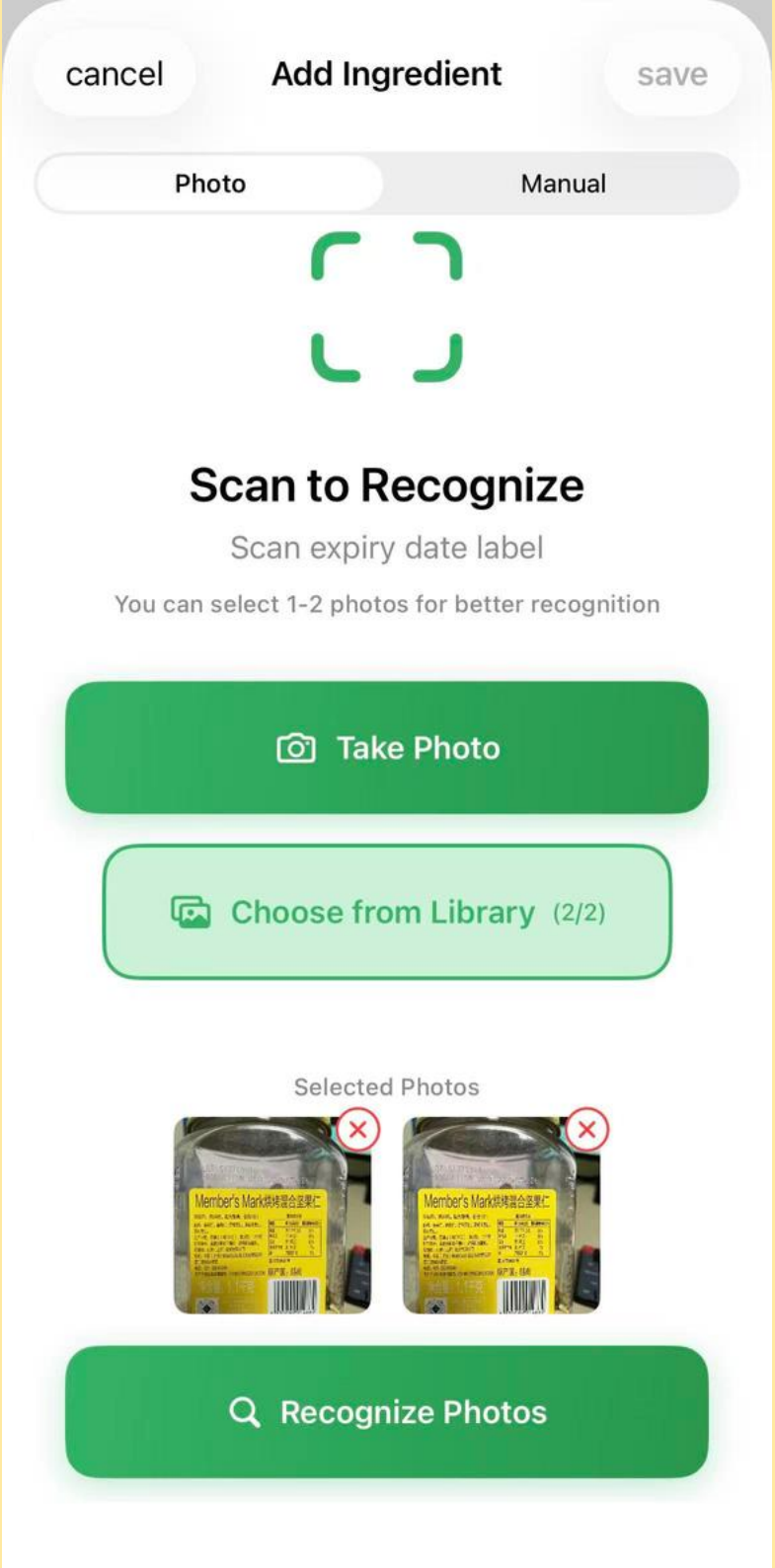


# ios-end Interface

## main page



## add & edit ingredients, set notification





The prototype has **strong potential** for real product we could use.

### **Pain Point Fit**

reducing food waste from expired ingredients + simplifying meal planning – aligns with urban/lifestyle needs.

### **Technical Feasibility**

Relies on mature 3rd-party APIs (ML Kit, AI recognition) + lightweight architecture; no technical bottlenecks, low development costs.

But....There are some similar products in real world, so we could deeply develop some **different functions**.

- Expand AI recognition for rare ingredients
- Optimize offline functionality
- Refine personalization with user feedback







# Thank You

**FoodieClock:  
Smart Fridge  
Companion**





# APPENDIX

**Web-end:** <https://foodieclock.vercel.app>

**Video-foodiclock–** <https://b23.tv/UzDdYAI>